

Sean Kerr | Solar Design Portfolio

Design Summary for Residential SLD Portfolio Item 1:

This residential concept design is built around a target ~20 kWh/day household load, using a conservative winter design insolation basis of ~3 peak sun hours per day during the critical 9:00–15:00 window in eastern Ontario. The PV array consists of two strings of ten 335 W optimized modules (~6.7 kW DC), connected to a 7.6 kW hybrid inverter with module-level shutdown via optimizers.

The battery system is sized for high autonomy, targeting approximately two days of full-load backup. In this iteration, the ESS consists of six LG RESU10 LV units for a total of ~58.8 kWh nominal at 48 V, ~1230 Ah capacity, providing roughly ~40–47 kWh usable energy assuming 80–90% depth of discharge and an overall system efficiency around 80%. This high-autonomy hybrid architecture is intended to support a ~20 kWh/day critical-load profile through multi-day winter outages while remaining compatible with Ontario net-metering and grid-interactive operation.